CHAPTER 22 · DERIVATIONS

SOUND WAVES

3 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 FUNDAMENTAL FREQUENCY: STRETCHED STRING, CLOSED PIPE AND OPEN PIPE

SYLLABUS 22.2

RESULT $f = \frac{v}{2L}$, $f = \frac{v}{4L}$, $f = \frac{v}{2L}$

1. Draw the fundamental standing wave for each system, then use length of the system = number of wavelengths in it, and $f = \frac{v}{\lambda}$.
2. **Stretched string** (length L, node at each fixed end, one antinode in the middle): $L = \frac{\lambda}{2}$, so $\lambda = 2L$ and $\mathbf{f = \frac{v}{2L} = \frac{1}{2L}\sqrt{\frac{T}{\mu}}}$, with $v = \sqrt{\frac{T}{\mu}}$.
3. **Closed pipe** (node at the closed end, antinode at the open end): $L = \frac{\lambda}{4}$, so $\lambda = 4L$ and $\mathbf{f = \frac{v}{4L}}$.
4. **Open pipe** (antinode at each end, one node in the middle): $L = \frac{\lambda}{2}$, so $\lambda = 2L$ and $\mathbf{f = \frac{v}{2L}}$.

> End correction is ignored. Overtones: string and open pipe give all harmonics ($f_n = \frac{nv}{2L}$); a closed pipe gives odd harmonics only ($f_n = \frac{(2n-1)v}{4L}$).

2 BEAT FREQUENCY

SYLLABUS 22.4(B)

RESULT $f = f_1 - f_2$

1. At a point: $y_1 = A \sin 2\pi f_1 t$ and $y_2 = A \sin 2\pi f_2 t$.
2. $y = y_1 + y_2 = 2A \cos(2\pi \frac{f_1 - f_2}{2} t) \sin(2\pi \frac{f_1 + f_2}{2} t)$.
3. The resultant sound has the average frequency $\frac{f_1 + f_2}{2}$ and a slowly varying amplitude $A_R = 2A \cos(\pi(f_1 - f_2)t)$.
4. The amplitude is maximum ($|A_R| = 2A$, loud sound) when $\cos(\pi(f_1 - f_2)t) = \pm 1$, i.e. $\pi(f_1 - f_2)t = 0, \pi, 2\pi, \ldots$ so $t = 0, \frac{1}{f_1 - f_2}, \frac{2}{f_1 - f_2}, \ldots$
5. The time between two successive loud sounds (period of the beat) is $T = \frac{1}{f_1 - f_2}$.
6. Beat frequency = $\frac{1}{T}$: $\mathbf{f = f_1 - f_2}$.

> The syllabus notes that this derivation is required.

3 DOPPLER EFFECT (FOUR CASES)

SYLLABUS 22.5(B)

RESULT $f' = \dfrac{f(v \pm u_o)}{v \mp u_s}$

1. Stationary source of frequency f; speed of sound v; wavelength $\lambda = \dfrac{v}{f}$.

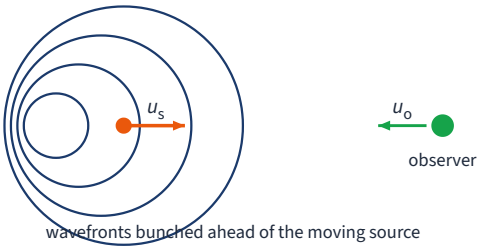
2. **(a) Observer moves towards the source** at u_o: the wavelength is unchanged, but the sound passes the observer at relative speed $v + u_o$, so $f' = \dfrac{v + u_o}{\lambda} = \mathbf{\dfrac{f(v + u_o)}{v}}$.

3. **(b) Observer moves away from the source**: relative speed $v - u_o$, so $\mathbf{f' = \dfrac{f(v - u_o)}{v}}$.

4. **(c) Source moves towards the observer** at u_s: in one period $\dfrac{1}{f}$ the wave moves $\dfrac{v}{f}$ while the source moves $\dfrac{u_s}{f}$, so the wavelength in front is shortened to $\lambda' = \dfrac{v - u_s}{f}$. The speed of sound is unchanged: $f' = \dfrac{v}{\lambda'} = \mathbf{\dfrac{fv}{v - u_s}}$.

5. **(d) Source moves away from the observer**: $\lambda' = \dfrac{v + u_s}{f}$, so $\mathbf{f' = \dfrac{fv}{v + u_s}}$.

6. Both moving: combine the two effects, $\mathbf{f' = \dfrac{f(v \pm u_o)}{v \mp u_s}}$ (upper signs when approaching).



wavefronts bunched ahead of the moving source

Observer moving: the wavelength stays the same, the relative speed changes. Source moving: the wavelength changes, the speed of sound stays the same.